In Person Public Health Training

# Interpreting Hepatitis B Serological Laboratory Results: Handout

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# Viral Hepatitis B, Quick Reference

## Purpose

This document was designed for providers and other hepatitis B related staff to use as a quick reference to review testing, clinical information, and other relevant viral hepatitis B information to support their patients and decision making.

## Testing

| Term | Definition |
|---|---|
| Screening | Testing people with no symptoms (asymptomatic) to find possible infection or immunity (one time CDC recommended adult triple panel test) |
| Testing (Diagnostic) | Testing people with symptoms (symptomatic), a plausible exposure, or prior positive result to confirm infection |
| Antigen | A non-self molecule that human immune systems use to identify pathogens (HBsAg, HBeAg) |
| Antibody | A protein produced by the human immune system to attack pathogens (Anti-HBs, Anti-HBc, Anti-HBe) |
| Qualitative | Laboratory result reported as a positive/negative or reactive/non-reactive |
| Quantitative | Laboratory result reported as a numeric value (anti-HBs>=10 mIU/mL, 2,000 HBV DNA IU/mL) |
| Detectable | A qualitative measure of disease antigen being present at the limit of a laboratory test or over |
| Undetectable | A qualitative measure of disease antigen being undetected by the laboratory test |

**Infection and Immunity**

| Term | Definition |
|---|---|
| Acute | Previously not infected with HBV, a newly acquired infection within the first six months of exposure |
| Chronic | HBsAg and or HBV DNA persisting over six months after acute status |
| Spontaneous Resolution | Immune system clears HBV without supportive treatment and develops protective antibodies |
| Immune due to natural infection | Protection after a previous infection, resolved due to supportive therapy or spontaneous resolution |
| Immune due to vaccination | Protection after HBV vaccine series developing protective antibodies anti-HBs |
| Reactivation | A flare up of someone with a chronic or resolved infection previously suppressed, often seen those in immunosuppressive therapy. |

**Laboratory Tests**

| Term | Definition |
|---|---|
| HBsAg | HBV surface antigen protein on the surface of the virus |
| Anti-HBs | Antibody to surface antigen, made by human immune system |
| Total IgM/IgG Anti-HBc | All immunoglobulin (antibodies) to the viral antigen core |
| IgM Anti-HBc | Core antibody mark of acute case within first six months of exposure |
| HBeAg | HBV e-antigen protein an indicator of active viral replication |
| Anti-HBe (HBeAb) | Antibody to the e-antigen, made by the human immune system |
| HBV DNA | The genetic material of HBV, used quantitatively for a viral load |
| Triple Panel | HBsAg, Anti-HBs, Total IgM/IgG Anti-HBc |

**Clinical Information**

| Term | Definition |
|---|---|
| Jaundice | Yellowing of the skin or eyes, can also be detected in urine and stool |
| Bilirubin | A breakdown of red blood cells processed by liver, Bilirubin>=3.0mg/dL for acute criteria |
| ALT, Alanine Aminotransferase | An enzyme found primarily in the liver, released when liver cells are damaged, ALT> 200 IU/L |
| AST, Aspartate Aminotransferase | An enzyme found around the body and in the liver, identified in liver damage but not diagnostic |
| HBIG, Hepatitis B Immunoglobulin | External hepatitis B immunoglobulin (antibody) given after exposure typically used for infants born to hepatitis B positive mother and also to people exposed via needlestick or sexual contacts |

**Surveillance Terms and Priority Populations**

| Term | Definition |
|---|---|
| Reportable disease | A disease that by law is required to be reported to the state health department, any positive hepatitis B laboratory results are required to be sent to Iowa HHS |
| Confirmed case | Meets the confirmatory laboratory criteria (and clinical criteria for acute case) from the 2024 CDC case definitions[1] |
| Probable case | Meets presumptive laboratory criteria (acute with clinical criteria) |
| Not a case | Does not meet confirmatory or presumptive laboratory criteria |
| PWID | People who inject drugs, person-first language for a priority population with higher exposure risk to hepatitis B |
| MSM | Men who have sex with men, a priority population with higher exposure risk to hepatitis B |
| Country of Origin | Identifies if a patient is US-born or if from a country with a higher prevalence of hepatitis B, helps determine likelihood of chronic status |

**CDC 2024 Case Classification[1]**

| Classification | Laboratory Evidence | Clinical Criteria |
|---|---|---|
| Acute, Confirmed Tier 1 | Positive HBsAg, HBeAg, or HBV DNA with IgM Anti-HBc or seroconversion within 12 months of negative HBsAg | No |
| Acute, Confirmed Tier 2 | Positive HBsAg or HBV DNA with IgM Anti-HBc not done | Yes |
| Acute, Probable | Positive IgM Anti-HBc with HBsAg, HBeAg and HBV DNA negative or not done | Yes |
| Chronic, Confirmed | Positive HBsAg/HBeAg twice six months apart or HBsAg/HBeAg plus Total IgM/IgG Anti-HBc or HBsAg plus HBeAg or HBV DNA | No |
| Chronic, Probable | HBsAg or HBeAg with negative IgM Anti-HBc or Not done | No |

**Interpretation of Laboratory Results[2]**

| HBsAg | Total Anti-HBc | IgM Anti-HBc | Anti-HBs | Interpretation |
|---|---|---|---|---|
| - | - | - | - | Never infected, susceptible, vaccinate |
| + | - | - | - | Early acute or false positive up to 18 days after vaccination |
| + | + | + | - | Acute infection |
| - | + | + | - | Acute infection, resolving |
| - | + | - | + | Recovered from past infection, immune |
| + | + | - | - | Chronic infection |
| - | + | - | - | Isolated core, past infection, low level chronic, or false positive |
| - | - | - | + | Immune from vaccination (Anti-HBs>=10mIU/mL) |

## References


1. Hepatitis B, acute and chronic 2024 Case Definition | CDC. May 30, 2024. Accessed November 6, 2026. https://ndc.services.cdc.gov/case-definitions/hepatitis-b-acute-and-chronic-2024/

2. Table 5 - STI Treatment Guidelines. July 15, 2021. Accessed November 6, 2026. https://www.cdc.gov/std/treatment-guidelines/table-5.htm